

Adaptive Median Filtering and Intuitionistic Fuzzy Learning with CNN-HTC Cascade R-CNN for Air Pollution Analysis

1st S Saran

Department of Electronics and
Communication Engineering
K S Rangasamy College of Technology
Tamil Nadu, India
sarans6363@gmail.com

2nd K M Siva kumar

Department
Department of Electronics and
Communication Engineering
K S Rangasamy College of Technology
Tamil Nadu, India
sivakumarm464@gmail.com

3rd M Devaki

Department of Electronics and
Communication Engineering
K S Rangasamy College of Technology
Tamil Nadu, India
devaki@ksrct.ac.in

Abstract— Air pollution has emerged as a critical environmental and public health challenge, demanding accurate and intelligent monitoring systems. This work proposes an advanced air pollution detection and classification framework integrating adaptive filtering, fuzzy segmentation, deep feature extraction, and robust object-based classification. Initially, raw air quality sensor data and pollution map images are pre-processed using Adaptive Median Filtering to effectively suppress impulse noise while preserving significant spatial and statistical patterns. Pre-processed image outputs are then processed through a Convolutional Neural Network (CNN) to extract high-level discriminative features representing pollutant intensity, spatial dispersion, and texture characteristics. These deep features are finally classified using a Hybrid Task Cascade (HTC) Cascade R-CNN, enabling precise multi-stage pollution category identification and severity estimation. The proposed framework supports multi-scale learning, contextual refinement, and improved generalization under varying environmental conditions. Experimental evaluation demonstrates enhanced detection accuracy, reduced false alarms, and robust performance under noisy and complex atmospheric scenarios. The integration of fuzzy logic with deep learning significantly improves boundary localization and classification reliability. Overall, the proposed system offers a scalable and intelligent solution for real-time air pollution monitoring and decision support in smart environmental management systems.

Keywords— Air Pollution Detection; Adaptive Median Filtering; KIFKM; Convolutional Neural Network; HTC Cascade R-CNN; Fuzzy Segmentation; Deep Learning; Environmental Monitoring; Intelligent Classification; Air Quality Assessment

I. INTRODUCTION

Air pollution has become one of the most pressing environmental concerns worldwide due to its direct impact on human health, climate change, and ecosystem sustainability. Rapid urbanization, industrial expansion, and increased vehicular emissions have significantly intensified the concentration of harmful pollutants in the atmosphere, necessitating reliable and intelligent air quality monitoring systems. Traditional air pollution analysis methods primarily rely on statistical models and threshold-based techniques, which often fail to capture complex spatial variations, nonlinear pollutant interactions, and uncertainty present in real-world environments. As a result, these conventional approaches exhibit limited accuracy and poor adaptability under dynamic atmospheric conditions. Recent advances in machine learning and deep learning have

opened new possibilities for automated and high-precision air pollution analysis.

However, raw sensor data and pollution map images are often contaminated by noise, missing values, and ambiguous boundaries, which adversely affect detection and classification performance. Effective pre-processing and uncertainty-aware segmentation are therefore crucial for enhancing model robustness. Fuzzy logic-based learning frameworks, particularly intuitionistic fuzzy models, have demonstrated strong capability in handling ambiguity and overlapping pollution regions.

In parallel, convolutional neural networks (CNNs) have proven highly effective in extracting discriminative spatial and contextual features from complex environmental data. Yet, single-stage classifiers may struggle with multi-scale pollution patterns and fine-grained severity estimation. To address these challenges, cascade-based deep architecture such as Hybrid Task Cascade (HTC) with Cascade R-CNN offer improved refinement through multi-stage learning and contextual reasoning. Motivated by these observations, this work introduces an integrated framework combining Adaptive Median Filtering, Kernel-based Intuitionistic Fuzzy learning, CNN-based feature extraction, and HTC Cascade R-CNN classification. The proposed approach aims to enhance noise resilience, segmentation accuracy, and classification reliability for comprehensive air pollution analysis. By synergizing fuzzy logic with deep learning, the framework provides a robust and scalable solution for intelligent air quality assessment and smart environmental monitoring systems.

II. LITERATURE SURVEY

[1] Armin Nakhjiri (2024): This paper represents a comprehensive analysis and predictive modeling framework for air pollution in industrial clusters, integrating multi-source data from satellite, ground monitoring, and meteorological sources. [2] Ruben Lévy (2025): This paper represents a statistical analysis inspecting the association between short-term exposure to key air pollutants and the significant peak in Acute Myocardial Infarction (AMI) mortality recorded on Mondays. [3] Alessandro Palma (2025): This study represents an empirical investigation examining the causal link between ambient air pollution exposure and increased workplace accident rates. [4] Yun-Jiu Cheng (2025): This paper presents an investigation into the combined impact of high-risk genetic profiles and

ambient air pollution exposure on cardiovascular disease incidence. [5] Mao Xu (2025): This paper represents a detailed analysis of PM2.5 pollution causes and concentrations within a key tropical forest ecosystem, highpoint the dual impact of seasonal forest fires and urban anthropogenic emissions.

[6] Christina Berninger (2025): This paper represents a longitudinal analysis inspecting the link between prenatal and early-life exposure to ambient air pollutants and adverse neurodevelopmental outcomes in young children. [7] Peng Shen (2025): This paper represents a behavioural analysis exploring how direct exposure to unembellished air pollution events influences real-time online donations to related environmental and health charities. [8] Dou Wang (2025): This paper represents a comprehensive review of evolving techniques for assessing individual-level air pollution exposure, highpoint the transformative potential of low-cost sensors, big data integration, and machine learning models. [9] Yueli Yao (2025): This paper represents a biomarker-based analysis demonstrating that chronic exposure to traffic-related air pollutants accelerates biological aging as measured by DNA methylation. [10] Shuangyu Wei (2025): This paper represents a field learning employing portable sensors to measure and analyse the personal air pollution exposure profiles of schoolchildren in a rural Welsh environment. The paper helps as a vital resource for considerate children's environmental health risks, informing local air quality policy, and promoting community-level public health interventions. [11] R Dayan (2026): This study presents a retrospective cohort analysis investigative the association between ambient urban air pollution exposure and key clinical outcomes of in vitro fertilization (IVF) cycles. [12] Qi Zhao (2025): This study employs a quasi-experimental design to estimate the causal impact of air quality improvements driven by environmental regulations on cognitive health outcomes in middle-aged and older adults in China. [13] Youcef Shahali (2026): This study presents a systematic meta-analysis synthesizing global evidence on the relationship between ambient air pollutants and airborne pollen focuses and allergenicity. [14] Valentina Cortesib (2025): This study presents a retrospective cohort analysis investigating the association between long-term ambient air pollution exposure and survival results in patients with (SCLC) and Large Cell Neuroendocrine Carcinoma (LCNEC). [15] Zihang Zhong (2025): This study employs a trajectory analysis of prospective cohort data to model the dynamic, long-term relationships between varying designs of air pollution exposure and the progression of concurrent respiratory and cardiovascular diseases.

III. PROPOSED METHODOLOGY

The proposed air pollution analysis framework is designed as a multi-stage intelligent system that integrates adaptive pre-processing, intuitionistic fuzzy learning, deep feature extraction, and cascade-based classification to achieve accurate and reliable pollution detection. The overall architecture consists of four major phases: data acquisition and pre-processing, fuzzy-based segmentation, deep feature extraction, and multi-stage classification. Initially, raw air quality sensor readings and pollution map

images are collected from monitoring stations and remote sensing sources. Since environmental data are often affected by impulse noise, sensor irregularities, and atmospheric disturbances, Adaptive Median Filtering (AMF) is employed as the pre-processing step. AMF dynamically adjusts the window size to suppress salt-and-pepper noise while preserving important spatial structures, edges, and statistical characteristics of pollution patterns. This step ensures noise-resilient input data for subsequent processing stages.

Following pre-processing, Kernel-based Intuitionistic Fuzzy K-Means (KIFKM) segmentation is applied to identify pollution regions accurately. KIFKM effectively manages uncertainty by incorporating both membership and non-membership functions, enabling robust handling of overlapping pollutant boundaries and non-linear data distributions. Once pre-processing is completed, the noise reduced image regions are apply to Convolutional Neural Network (CNN) for deep feature extraction. The CNN automatically learns high-level discriminative features such as pollutant intensity variations, spatial dispersion, texture information, and contextual relationships. These features capture both local and global pollution characteristics, facilitating precise representation of air quality patterns across multiple scales. Finally, the extracted deep features are classified using a Hybrid Task Cascade (HTC) Cascade R-CNN framework.

Overall, the integration of adaptive filtering, intuitionistic fuzzy learning, CNN-based feature extraction, and HTC Cascade R-CNN classification enables a robust, scalable, and intelligent air pollution monitoring system. This multi-stage architecture performs progressive refinement of detection and classification outputs, enabling accurate pollution category identification and severity estimation (Fig 1).

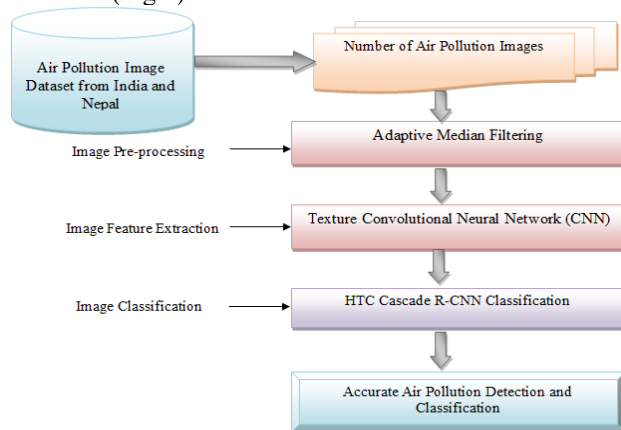


Figure 1. Architecture Diagram of Adaptive Median Filtering and CNN-HTC Cascade R-CNN for Air Pollution Analysis

The cascade structure improves boundary localization and reduces false alarms through iterative contextual learning and feature enhancement. The proposed methodology effectively addresses noise, uncertainty, and complex atmospheric variations, making it suitable for real-time environmental monitoring and smart decision-support applications.

A. Pre-processing: Adaptive Median Filtering

Important stage in air pollution analysis to improve the quality of images obtained from satellite and ground-based monitoring systems. Air pollution images are frequently unnatural by impulse noise due to atmospheric disturbances, sensor imperfections, and data transmission errors. Adaptive Median Filtering is an effective non-linear filtering method used to suppress such noise while preserving essential image specifics. The algorithm functions by examining each pixel within a local neighbourhood window. Initially, a minor window size is designated nearby the target pixel. The least, extreme, and median intensity values within the window are computed. These statistical values help determining whether the current pixel is noisy or noise-free.

Adaptive Neighbourhood Window Formation

Adaptive neighbourhood window filtering achieves well under varying noise conditions. The filtered image exhibits developed clarity and contrast and describe following equation

$$W_{xy}(S) = \{I(i,j) \mid |x-s| \leq x+s, y-s \leq y+s\} \dots \text{Eq 3.1}$$

Where above equation $I(i,j)$ represents the pixel intensity at location (i,j) , (x,y) denotes the coordinates of the current pixel under consideration, s indicates the adaptive window radius, and (s) refers to the local neighbourhood window centered at (x,y) .

If the median value lies between the least and extreme values, the window is considered suitable. If this disorder is not satisfied, the window size is improved adaptively. This adaptive expansion carry on until a valid window is obtained or a predefined extreme size is touched. Once a suitable window is recognized, the procedure evaluates the center pixel. If the pixel value lies within the local intensity range, it is reserved.

Median Intensity Computation

$$Z_{med} = \text{median}(W_{xy}(S)) \dots \text{Eq 3.2}$$

Where above equation represent the Z_{med} represents the median pixel intensity, $\text{median}(\cdot)$ denotes the median operator, and $W_{xy}(s)$ refers to the adaptive neighbourhood window.

The noisy pixel is then exchanged by the median value of the window. This selective filtering procedure prevents needless smoothing of clean pixels. As a result, essential structures such as pollution boundaries and smoke dispersion designs are conserved.

Minimum and Maximum Intensity Computation

$$Z_{min} = \min(W_{xy}(S)) \dots \text{Eq 3.3}$$

$$Z_{max} = \max(W_{xy}(S)) \dots \text{Eq 3.4}$$

Where Z_{min} represents the minimum pixel intensity in the window, $W_{xy}(s)$ denotes the adaptive neighbourhood window, Z_{max} represents the maximum

pixel intensity in the window, and $W_{xy}(s)$ refers to the adaptive neighbourhood window.

The technique effectively eliminates high-density salt-and-pepper noise. The pre-processed image develops segmentation performance. Feature extraction develops extra reliable after noise removal. Classification accuracy is improved due to improved image quality. The procedure is computationally efficient. It is suitable for large-scale environmental monitoring. Adaptive Median Filtering thus plays a significant role in air pollution image pre-processing.

B. Feature Extraction: Texture Convolutional Neural Network (CNN)

Texture convolutional Neural Networks are extensively used for automatic feature extract due to their hierarchical learning capability. Texture CNN architecture contains of convolutional, activation, and pooling layers. The convolution level applies learnable mesh to the input image. These filters detect low-level features such as edges and textures. In air pollution images, edges correspond to smoke boundaries and pollution regions. The convolution operation generates multiple features maps.

Activation Function (ReLU)

$$A_k(x,y) = \max(0, F_k(x,y)) \dots \text{Eq 3.5}$$

Where $A_k(x,y)$ represents the activated feature map, $F_k(x,y)$ denotes the convolution output, and $\max(\cdot)$ represents the Rectified Linear Unit (ReLU).

Activation functions present non-linearity to the extracted features. Rectified Linear Unit is usually used to improve learning efficiency. Pooling layers decrease the spatial dimensions of feature maps. This reduction increases computational effectiveness and robustness to noise. Max pooling preserves dominant pollution features. Deeper convolution layers capture complex designs such as dense pollution clusters. Feature hierarchies allow discrimination between polluted and non-polluted regions.

Feature Map Flattening

$$f = \text{Flatten}(P_k) \dots \text{Eq 3.6}$$

Where P_k represents the pooled feature maps, f denotes the one-dimensional feature vector, and $\text{Flatten}(\cdot)$ represents the vectorization operation.

The extract feature maps are flattened into feature vectors. These vectors characterize spatial and semantic data of the image. The texture CNN automatically learns discriminative features without manual design. Feature extraction is optimized through back propagation. The learned features adapt to varying pollution intensities.

Texture Connected Layer Transformation

$$z = WfT + bf \dots \text{Eq 3.7}$$

Where W_f represents the weight matrix of the fully connected layer, b_f denotes the bias vector, T represents the texture feature vector, and z denotes the extracted high-level features.

Texture CNN features are robust to illumination variations and background noise. This robustness is crucial for outdoor air pollution monitoring. The extracted features improves segmentation accuracy. They also enhance classification and prediction tasks. Texture CNN-based features support real-time air quality assessment. The technique decreases dependency on handcrafted features. It increases generalization across different atmospheres. Texture CNN feature extraction thus plays a vital role in air pollution image analysis.

C. Classification: HTC Cascade R-CNN Classification

Image classification is a essential task in air pollution analysis for categorizing pollution levels and source from environmental images. Air pollution images captured through satellite sensors and surveillance cameras contain complex visual patterns such as smoke plumes, industrial emissions and haze. Hybrid Task Cascade (HTC) Cascade R-CNN is a strongest deep learning framework planned for accurate object classification and detection. The HTC framework assimilates detection and segmentation responsibilities in a cascaded manner. It forms upon the Region-based Convolutional Neural Network architecture. The procedure begins with a backbone network that excerpts deep feature representations.

Feature Pyramid Networks are used to capture multi-scale pollution topographies. Region Proposal Networks generate candidate regions of interest. These regions potentially agree to pollution sources or dense emission areas. The cascade building refines proposals through several stages. Every stage applies progressively stricter classification thresholds. This improves discrimination between pollution categories.

$$F=\phi(I).....Eq\ 3.8$$

Where I represents the input air pollution image, $(\cdot)$ denotes the cascade CNN (e.g., ResNet), and F represents the extracted deep feature maps.

Bounding box regression enhances localization accuracy. Semantic information is shared across cascade stages. This hybrid task design jointly optimizes classification and segmentation. This joint learning improves contextual understanding of pollution regions. The model effectively distinguishes smoke, haze and background regions.

HTC decreases untrue positives in complex atmospheric conditions. High-level features capture fine-grained pollution appearances. The multi-stage refinement increases classification confidence. The architecture handlebars overlapping pollution regions effectually.

Feature sharing improves computational efficiency. The prototypical adapts well to changing illumination and weather conditions. It offers robust presentation for outdoor monitoring. The confidential outputs support pollution level assessment. They also aid in source identification. The method develops decision-making in environmental management. HTC Cascade R-CNN thus offers reliable air pollution image classification.

IV. Experimental Evaluation

The experimental evaluation of the proposed air pollution analysis framework was conducted using the Air Pollution Image Dataset from India and Nepal, which contains real-world pollution images captured under diverse environmental and atmospheric conditions. The dataset includes images representing varying pollution levels, such as low, moderate, and severe air quality scenarios, collected from multiple urban and semi-urban regions. This diversity ensures robust evaluation across different spatial patterns, lighting conditions, and pollution intensities.

Dataset Name

Air Pollution Image Dataset from India and Nepal

Url

<https://www.kaggle.com/datasets/adarshrrouniyar/air-pollution-image-dataset-from-india-and-nepal?resource=download>

V. Result and Discussion

A. Accuracy

Accuracy is defined as the proportion of the correct predictions (both total no of images and predict correct images) presented by the detection system out of every prediction (Fig 2). Accuracy is mathematically calculated below following equation,

$$\text{Prediction Accuracy (\%)} = \frac{\text{No Of Correct predict images}}{\text{Total No of images}} * 100 \dots \text{Equ 5.1}$$

Proposed technique (HTC-RCNN) higher accuracy when compared with Existing 1 VCD [1] and Existing 2 AMI[2] technique

Table I Prediction Accuracy of HTC Cascade R-CNN

Number of Input images	Prediction Accuracy (%)		
	HTC-RCNN	Vertical Column Density(VCD)	Acute myocardial Infarction(AMI)
10	97	91	88
20	95	92	84
30	97	94	81
40	94	90	84

50	95	93	85
60	96	92	83
70	94	91	89
80	96	93	87
90	97	95	89
100	95	92	90

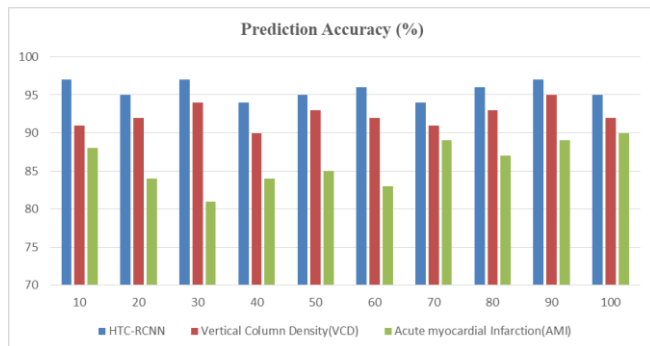


Figure 2 . Prediction Accuracy(%) Comparison

B. Error Rate (%)

Error rate is defined (Fig 3) as the ratio of incorrect predictions made by the model out of the total number of cases,

$$\text{Error rate} = (\text{Fp} + \text{FN}) / (\text{Tp} + \text{TN} + \text{FP} + \text{FN}) \dots \text{Equ 5.2}$$

Where above equation 5.2 represent FP –false positive rate, FN-False negative rate, TP-True positive rate, TN-True negative rate,

Table II Error Rate of HTC Cascade R-CNN

Number of Input Images	Error rate (%)		
	HTC-RCNN	Vertical Column Density(VCD)	Acute myocardial Infarction(AMI)
10	6	11	14
20	9	13	16
30	12	18	17
40	15	22	22
50	23	26	31
60	27	30	36
70	28	33	38
80	34	41	43
90	36	45	55
100	38	57	59

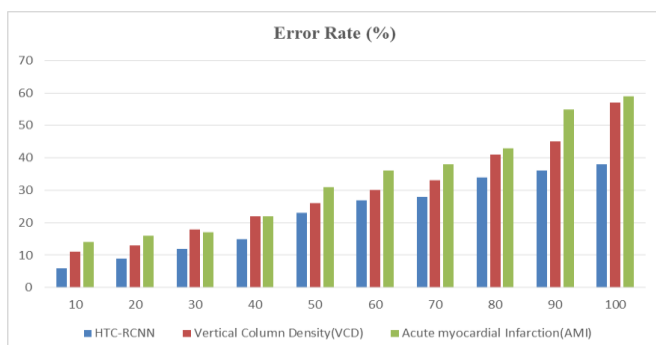


Figure 3 . Error Rate(%) Comparison

model outperforms Proposed technique (HTC-RCNN) higher accuracy when compared with Existing 1 VCD [1] and Existing 2 AMI[2] technique in terms of prediction accuracy and error rate. This leads into greater dependability and more accurate decision-making in website phishing detection, eventually surpassing the performance of previously used strategies.

C. Prediction Time (ms)

Prediction time is determined (Fig 4) as the number of time taken by the system to process the task from input to output. Additionally, it analyses the input and produce the prediction

$$\text{PT} = \text{Time at End of Prediction-Time at Start of}$$

$$\text{Prediction} \dots \text{Equ 5.3}$$

Table III Prediction time of HTC Cascade R-CNN

Number of Input Images	Prediction time (ms)		
	HTC-RCNN	Vertical Column Density(VCD)	Acute myocardial Infarction(AMI)
10	31	38	56
20	29	35	51
30	41	51	59
40	48	54	64
50	51	59	66
60	54	67	70
70	56	71	76
80	62	79	83
90	65	81	87
100	66	85	90

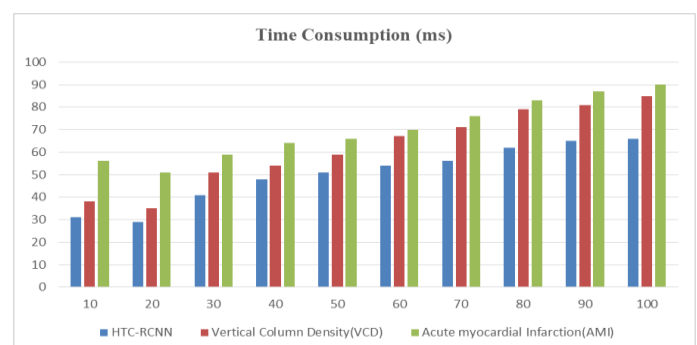


Figure 4 . Prediction Time(ms) Comparison

Produces faster computation and prediction processes. Compared to previous techniques, the RCNN built with this improved knowledge set completes the arrangement substantially faster, hence removing the absolute prediction time. The suggested technique Proposed technique (HTC-RCNN) higher accuracy when compared with Existing 1 VCD [1] and Existing 2 AMI[2] technique.

V. CONCLUSION

This study presented an intelligent and robust air pollution detection and classification framework that integrates Adaptive Median Filtering, Kernel-based Intuitionistic Fuzzy learning, CNN-based deep feature extraction, and HTC Cascade R-CNN classification. The proposed methodology effectively addresses key challenges in air pollution analysis, including noise contamination, uncertainty in pollutant boundaries, and complex spatial variations. Adaptive Median Filtering enhances data quality by suppressing impulse noise while preserving essential spatial patterns. The CNN model successfully extracts high-level discriminative features related to pollutant intensity, dispersion, and texture, enabling accurate representation of air quality patterns. The HTC Cascade R-CNN further refines classification through multi-stage learning, resulting in improved detection accuracy and reduced error rate.

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